Course Overview

Python for AI-Driven Automation and Business Data Science

Course materials

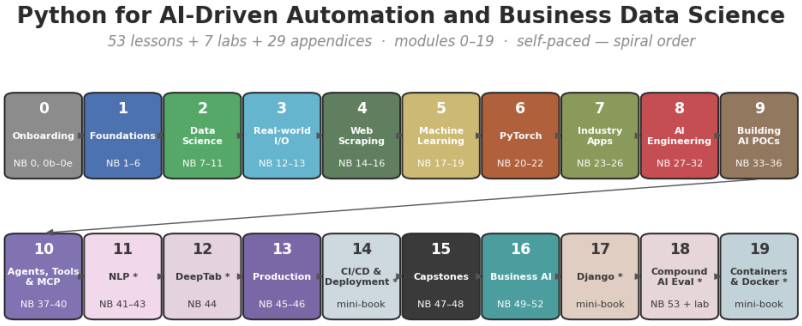
Contents

1. The big picture
2. Structure
3. Learning paths
4. How to study
5. What you'll build
6. Getting started

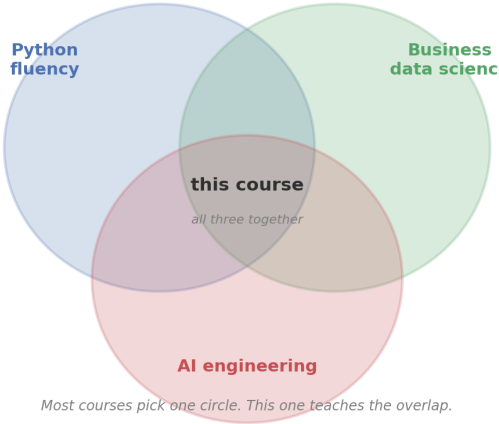
Outline

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The course in one diagram



Where this course sits



Most courses pick one circle. This one teaches the overlap.

Three skills, taught together

1. Python fluency

Read, write, and modify clean Python without friction. Functions, modules, error handling, defensive parsing.

2. Working with data professionally

pandas, NumPy, matplotlib, scikit-learn, statistics, time-series forecasting — plus PyTorch deep learning for when the classical toolkit stops. Real datasets, not toy ones.

3. AI-driven automation

Calling LLMs as functions. Structured output. Retrieval-augmented generation. Tools and agents. Document processing. *Plus* the engineering wrapper around all of it.

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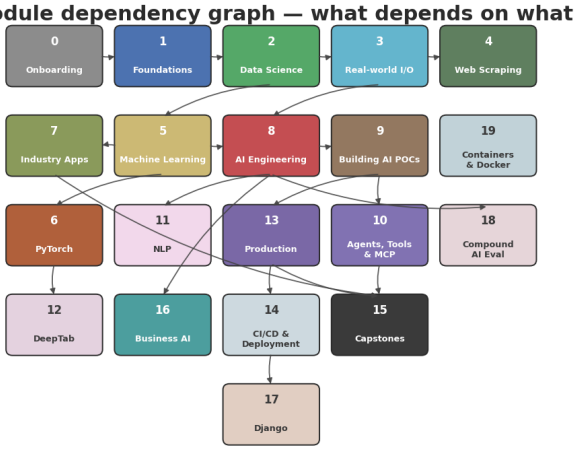
6 Getting started

Twenty modules — 53 lessons, 7 labs (+ 29 appendices)

#	Module	Notebooks
0	Onboarding	master onboarding, overview, see-it-work, command line, Git
1	Foundations	NB 1–6 — Python you can read without friction
2	Data science	NB 7–11 — pandas, NumPy, plots, stats, time series
3	Real-world I/O	NB 12–13 — HTTP, SQL, validation
4	Web scraping	NB 14–16 — BeautifulSoup, Firecrawl, the OpenAlex API
5	Machine learning	NB 17–19 — sklearn, model evaluation, features
6	Deep learning (PyTorch)	NB 20–22 — tensors & the training loop, training craft, embeddings & serving
7	Industry applications	NB 23–26 — churn value, fraud, segmentation, demand
8	AI engineering	NB 27–32 — LLM patterns, RAG, agents, evaluation
9	Building AI POCs	NB 33–36 — setup, three POCs, RAG deep dive, agentic AI
10	Agents, tools & MCP	NB 37–40 — architectures, robust tools, MCP, multi-agent
11	NLP (optional)	NB 41–43 — topic modeling, sentiment
12	DeepTab (optional)	NB 44 — deep learning for tabular data
13	Production	NB 45–46 — packaging, scheduling, config & secrets
14	CI/CD & deployment	mini-book + 3 labs — Docker, GitHub Actions, HTTPS
15	Capstones	NB 47–48 — analytics + AI assistant
16	Business AI	NB 49–52 — strategy, architecture, BPM, governance
17	Django (optional)	mini-book + 2 labs — serve a model behind a web UI
18	Compound AI eval (opt.)	NB 53 + lab — factorial designs, ANOVA attribution, CAFE
19	Containers (optional)	mini-book + lab — images, layers, namespaces, Dockerfiles

Every module folder has its own README — goals, prerequisites, what you'll build. Read it before the module's first notebook. Module and lesson numbers follow the learning order.

Module dependencies — what comes before what



Edges flow downstream. Module N is safe to start once every upstream module is comfortable.
Module 19 has no upstream: it needs only Module-1 Python and a terminal, and pairs with 14 in either order.

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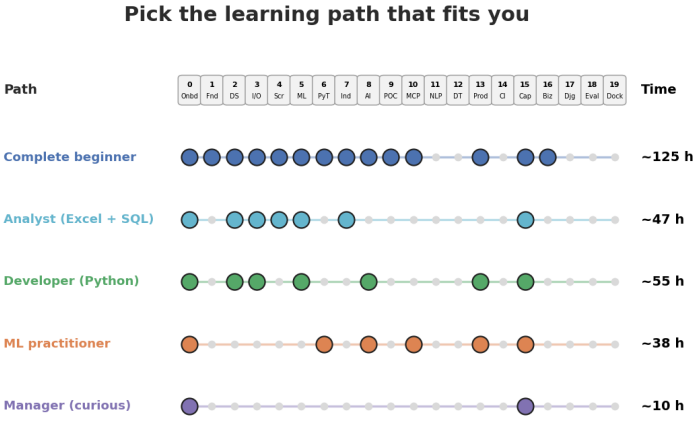
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Pick the path that fits you



Each dot is one module. Solid line = modules you'd touch on that path. Optional modules (11, 12, 14, 17, 18, 19) join any path.

When to pick which path

Complete beginner

Never written code before. Take all modules in **spiral order** (see it work first, then the skills, then the builds). Time-budget: **about 125 hours** — the interactive estimator in 00b computes yours.

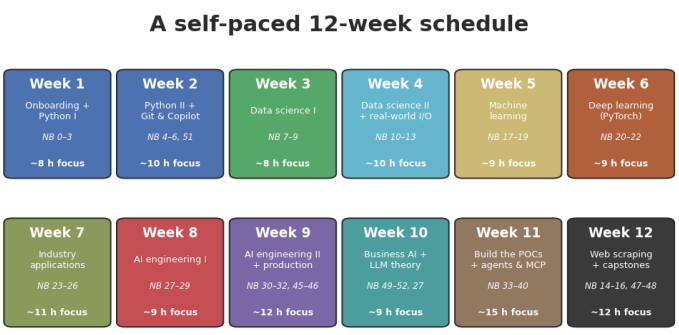
Analyst who knows Excel and SQL

Skim Module 1 (Python basics); Module 3's SQL will feel familiar. Focus on Modules 2 (data science), 4 (web scraping), 5 (ML), 7 (industry applications), 15 (capstone). Time-budget: **about 47 hours**.

ML practitioner adding AI engineering

Skip foundations and data science. Focus on Modules 6 (PyTorch), 8 (AI engineering), 10 (agents & MCP), 13 (production), 15 (capstone). Time-budget: **about 38 hours**.

Twelve weeks, self-paced



≈ 8–15 hours of focused study per week · more time → faster, less time → split a week in two

Got more time? Move faster. Got less? Split one week into two.

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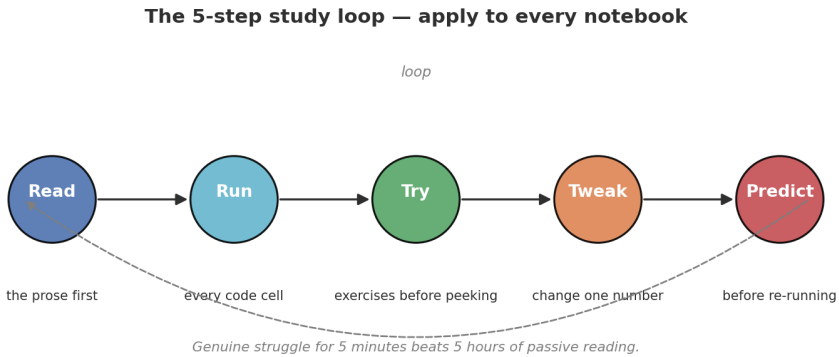
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The five-step study loop



Apply this loop to *every* notebook. Genuine struggle beats passive reading.

Eight visual markers you'll see throughout

Marker	Meaning
`(Tip)`	A small "stop and notice" callout.
`(Intuition)`	The mental model behind the syntax.
`(Pitfall)`	A bug or anti-pattern that bites if you're not warned.
`(Quick exercise)`	A ∼2-minute in-lesson checkpoint with a collapsible solution.
`(Exercises)`	Hands-on tasks. Try *before* peeking at the solution.
`(Bonus)`	One larger applied task per notebook.
`(Debug me)`	A cell that *intentionally* contains a bug for you to find.
`(Next step)`	Always points to the next notebook in your path.

Once you internalise the markers they become road signs, not decoration.

Checkpoints, quizzes, and the fast track

389 in-lesson checkpoints — the rhythm of every lesson

Roughly every ∼20 minutes of material: a `(Quick exercise)` with a scaffolded starter and a collapsible solution — every solution executed in a fresh kernel. Teach → try for ∼2 minutes → reveal.

17 module quizzes

Five multiple-choice questions per content module (∼10 minutes) in `quizzes/` — read code, predict behaviour, pick the right tool. Score 0–2/5? Re-do the module before moving on.

Short on time? The fast track

`fast_track/` condenses the whole course into 22 trimmed lessons (∼26 hours) — the 14 core essentials plus 8 breadth extensions.

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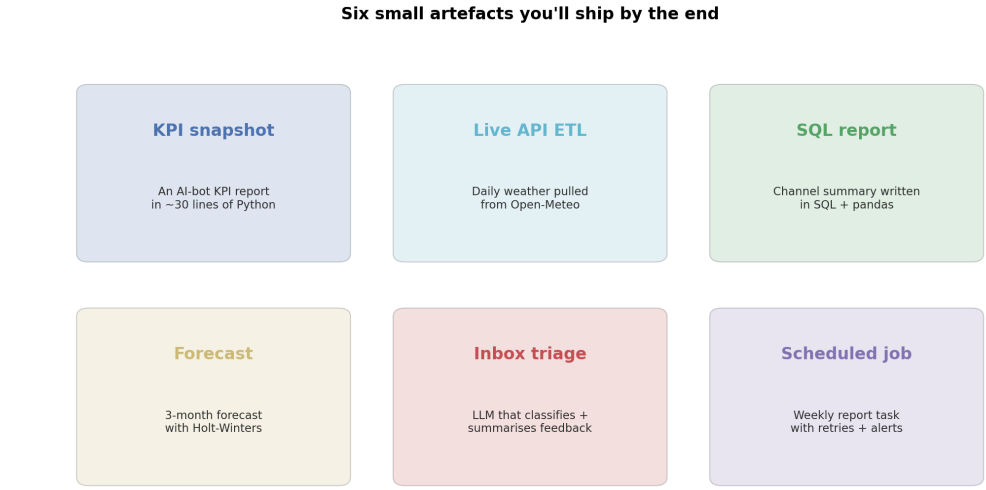
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Six small artefacts you ship along the way



Plus two big capstones at the end

Capstone A — AI Support-Bot Analytics (NB 47)

Five channels $\times$ twelve months of operational data $\rightarrow$ 2×2 executive dashboard $\rightarrow$ regression demonstrating Simpson's paradox $\rightarrow$ five-bullet executive summary.

Capstone B — AI Customer-Feedback Assistant (NB 48)

Ingest free-form customer feedback $\rightarrow$ classify $+$ extract via the MockLLM $\rightarrow$ ground answers in a knowledge base via RAG $\rightarrow$ orchestrate as a scheduled job with alerts. End-to-end AI feature.

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LLM providers — local and hosted, four options

Provider	Class	Best for
OpenAI	`OpenAILLM("gpt-5.4-mini")`	Reliable default
Anthropic	`AnthropicLLM("claude-haiku-4-5")`	Long context, careful tone
Google	`GoogleLLM("gemini-2.5-flash")`	Cheapest at high volume
Ollama	`OllamaLLM("llama3.2:3b")`	**Local** – no internet, no key

Default = the offline MockLLM

Every AI notebook runs offline by default. Swapping in a real provider is a *one-line* change thanks to the unified interface in `llm_providers.py` at the repo root.

See the dedicated guide

Notebook `08_ai_engineering/A1_llm_providers_guide.ipynb` has the full setup walk-through, model recommendations, and cost estimates.

What you need to install

For Google Colab

Nothing. All required libraries are pre-installed. Open the notebook, hit Shift+Enter.

For local Jupyter

```
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
jupyter lab
```

Notebook 0 includes an environment-check cell you run to confirm everything works before you move on.

Now → open Notebook 0

One file to open. The rest unfolds from there.

`00_onboarding/00_master_onboarding.ipynb`

Three things that notebook does for you

1. Confirms your Python environment is correctly set up.
2. Explains the five-step study loop you'll use throughout.
3. Helps you pick the right learning path from the five options.

One last reminder before you start

Five-step study loop, in your pocket

Read the prose first → **Run** every code cell → **Try** the exercises before peeking → **Tweak** one number → **Predict** the new output before re-running.

When to use which marker

`(Tip)` information; `(Intuition)` mental model; `(Pitfall)` avoid this; `(Quick exercise)` ∼2-minute checkpoint; `(Exercise)` hands-on practice; `(Bonus)` bigger applied task; `(Debug me)` intentional bug — find it.

When you're stuck for 15 minutes

Copy the full error into a search engine. 95% of beginner errors are well-documented. The course is designed to teach you to debug, not to spare you the experience.

Welcome.

134 notebooks await you.

Open Module 0 next.